Internship Assessment

A Report submitted in partial fulfillment of the requirement for the degree of

B.Tech

In

Computer Science and Engineering

REPORT WRITING

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DECLARATION

This is to certify that the Mini Project Report Entitled "Linux-Based DNS Server and Domain Hosting System" which is being submitted in partial fulfillment of the requirement for the award of degree B. Tech in Computer Science and Engineering to RKGIT, Ghaziabad (Dr. A.P.J. Abdul Kalam Technical University, Lucknow) comprises only original work and studies carried out by the students themselves. The matter embodied in this work has not been submitted for the award of any other degree**.**

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Abstract

The Linux-Based DNS Server and Domain Hosting System is a practical implementation of a network management solution designed to establish and manage domain name resolution and web hosting services using open-source technologies. The project aims to provide a cost-effective, secure, and scalable alternative to commercial hosting platforms by utilizing the flexibility and reliability of the Linux operating system.

The system leverages BIND9 (Berkeley Internet Name Domain) for DNS configuration and Apache Web Server for domain hosting, forming the backbone of the project's architecture. It enables the translation of human-readable domain names into IP addresses through forward and reverse lookup zones, ensuring seamless communication within the network. Additionally, it supports the hosting of multiple domains through virtual hosting, allowing efficient management of various websites from a single server.

The development process involved the configuration of DNS records, such as A, CNAME, MX, and PTR entries, and the implementation of both Primary (Master) and Secondary (Slave) DNS servers for redundancy and reliability. Tools like nslookup and dig were used to validate DNS functionality and ensure correct domain mapping. Security measures, including restricted zone transfers and firewall protection, were implemented to safeguard the system against external threats and unauthorized access.

This project demonstrates the significance of self-managed DNS and hosting infrastructures in enhancing organizational control and reducing dependency on third-party service providers. It serves as a comprehensive learning model for understanding network communication, server configuration, and system administration under a Linux environment. Overall, the Linux-Based DNS Server and Domain Hosting System contributes to the development of robust, secure, and efficient web hosting frameworks suitable for academic, institutional, and small-scale industrial applications.

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Introduction

The Summer Internship program conducted at Netcamp Solutions Private Limited, in association with Raj Kumar Goel Institute of Technology (RKGIT), Ghaziabad, from 4th July 2025 to 24th August 2025, provided students with the opportunity to gain hands-on experience in Linux Administration, Network Configuration, and Server Management. The training was designed to bridge the gap between academic learning and real-world technical application, focusing on the core domains of Web Hosting, DNS Configuration, and System Security.

During the internship, participants were introduced to various Linux-based server technologies, including DNS (Domain Name System) management, Apache web hosting, and domain mapping, all of which form the backbone of internet infrastructure. The internship emphasized practical exposure over theoretical knowledge, allowing students to deploy and manage live servers in a controlled network environment. This real-time implementation not only improved their technical proficiency but also enhanced their understanding of network protocols, IP addressing, and domain resolution mechanisms.

As part of this program, the project titled "Linux-Based DNS Server and Domain Hosting System" was undertaken. The primary objective of this project is to design and implement a secure, efficient, and scalable DNS infrastructure capable of resolving domain names within a private or public network. Additionally, the project focuses on hosting multiple domains using Linux-based tools and services, enabling organizations to manage websites and subdomains from a centralized server environment.

The system utilizes BIND (Berkeley Internet Name Domain) as the core DNS software on a Linux platform, which serves as an industry-standard tool for domain resolution. Apache Web Server is employed for hosting websites, while configuration files such as named.conf and zone files are customized to define forward and reverse lookup zones. The implementation also integrates network configuration, firewall settings, and service monitoring to ensure stability, performance, and security.

One of the major learning outcomes of this project was understanding how DNS translates human-readable domain names into IP addresses, facilitating seamless web access. Students learned to create and manage Primary (Master) and Secondary (Slave) DNS servers to ensure redundancy and high availability. Additionally,

domain hosting was implemented to allow multiple websites to be served from a single Linux machine using Virtual Hosting techniques.

The development and deployment process followed a structured approach inspired by the System Administration Lifecycle, including stages such as requirement analysis, system design, implementation, testing, and monitoring. Tools like Ubuntu Server, BIND9, and Apache2 were used, with configurations tested using nslookup, dig, and ping utilities to validate correct DNS behaviour. Through this project, students gained insight into server automation, troubleshooting, and security hardening, preparing them for professional roles in system administration and network engineering.

In addition to technical expertise, the internship promoted key professional skills such as collaboration, documentation, and problem-solving. Regular mentor interactions and practical demonstrations reinforced the importance of accuracy, attention to detail, and systematic debugging in network management.

In conclusion, the Linux-Based DNS Server and Domain Hosting System project provided a comprehensive learning experience by combining theoretical understanding of networking with hands-on server deployment. It highlighted the vital role of DNS and hosting technologies in modern communication systems and showcased how Linux-based solutions can deliver reliable, secure, and cost-effective network infrastructures. The internship experience thus served as a solid foundation for future endeavors in cybersecurity, system administration, and cloud computing.

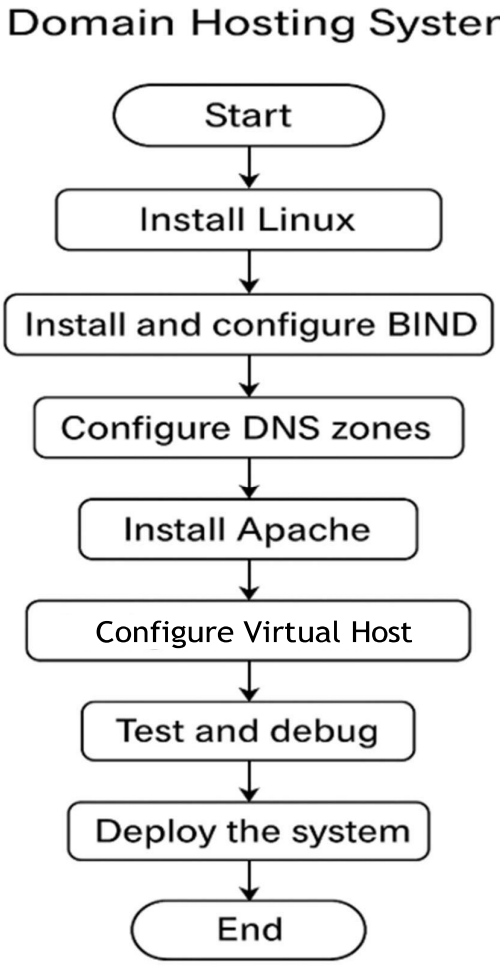


Fig.1.Flowchart

Problem Statement

In today's digital ecosystem, the Domain Name System (DNS) serves as a fundamental backbone of the Internet, translating human-readable domain names into machine-readable IP addresses. Every online service—whether a website, application, or email system—depends on DNS for proper communication and accessibility. However, despite its importance, many organizations face challenges in deploying, maintaining, and securing DNS and hosting infrastructures effectively, particularly when relying on third-party or commercial hosting providers.

The conventional approach of using cloud-based or managed DNS services often introduces issues such as limited control, subscription costs, and restricted customization. Small- and medium-scale enterprises, educational institutions, and independent developers frequently struggle with the complexity of configuring DNS environments or hosting multiple domains in a secure and scalable manner. Moreover, a lack of understanding of how DNS servers operate within a Linux environment limits opportunities for optimization, automation, and troubleshooting.

Another major challenge is ensuring domain hosting reliability and fault tolerance. Many hosting systems depend on centralized resources, making them vulnerable to failures, network outages, or security breaches. Without proper configuration of primary and secondary DNS servers, a network may experience downtime, resulting in inaccessibility of hosted websites. Additionally, improper zone configuration or lack of reverse mapping can lead to slower domain resolution, incorrect routing, or degraded user experience.

From a security perspective, open DNS servers are often exposed to spoofing, cache poisoning, and unauthorized zone transfers, which can compromise data integrity and lead to malicious redirection of traffic. Therefore, implementing a secure and self-managed DNS hosting solution becomes essential to protect organizational assets and ensure uninterrupted connectivity.

Solution Approach

The proposed solution is to develop a Linux-based system that integrates BIND9 DNS Server and Apache Web Server to create a complete domain hosting environment capable of resolving and serving multiple domains efficiently. This system emphasizes open-source technologies, offering a practical, low-cost, and flexible alternative to commercial hosting platforms.

1. System Design and Architecture

The system follows a client-server architecture, where the Linux server acts as both the DNS resolver and web host. The DNS service manages domain-to-IP mapping, while Apache handles web content delivery. The design includes:

- Primary DNS Server (Master): Responsible for hosting and maintaining authoritative zone files.
- Secondary DNS Server (Slave): Ensures redundancy by synchronizing records from the master.
- Web Server (Apache): Configured for virtual hosting to serve multiple websites from a single server.

2. Implementation Steps

- **Installation and Configuration:** Install Linux and configure BIND9 for DNS operations. Define forward and reverse lookup zones in named.conf and create zone files specifying records.
- **Apache Setup:** Install Apache2 and configure virtual hosts for each domain. Each domain is linked to its DNS record, allowing seamless name resolution and hosting.
- **Testing:** Verify DNS resolution using tools like dig, nslookup, and ping. Confirm that each domain correctly maps to its hosted website.
- **Security and Hardening:** Restrict zone transfers, enable firewalls (UFW/iptables), and disable recursion for unauthorized users to prevent external exploitation.

3. Features of the Proposed System

- Efficient domain-to-IP resolution with customizable DNS records.
- Virtual hosting capability for multiple domains.
- High availability through primary and secondary DNS configuration.
- Secure communication and restricted administrative access.
- Scalable architecture supporting future domain additions.

4. Testing and Validation

The system undergoes unit testing for individual modules (DNS setup, Apache configuration) followed by integration testing to ensure seamless interaction. Performance is validated by measuring query response time, uptime, and hosting accuracy. Stress testing verifies reliability under concurrent requests.

5. Future Scope

Future enhancements include:

- Integration with SSL/TLS certificates for HTTPS hosting.
- Automation using Shell or Python scripts for DNS updates.
- Implementation of monitoring tools like Nagios for uptime tracking.
- Extension to cloud or hybrid deployments for large-scale scalability.

Conclusion

The proposed Linux-Based DNS Server and Domain Hosting System offers a complete, open-source, and secure solution for managing web domains. By leveraging BIND and Apache on Linux, it ensures control, reliability, and efficiency—empowering users to host and manage multiple domains independently without dependency on external providers. This project not only addresses the existing limitations of commercial DNS services but also enhances technical understanding of server management, making it a valuable educational and industrial prototype.

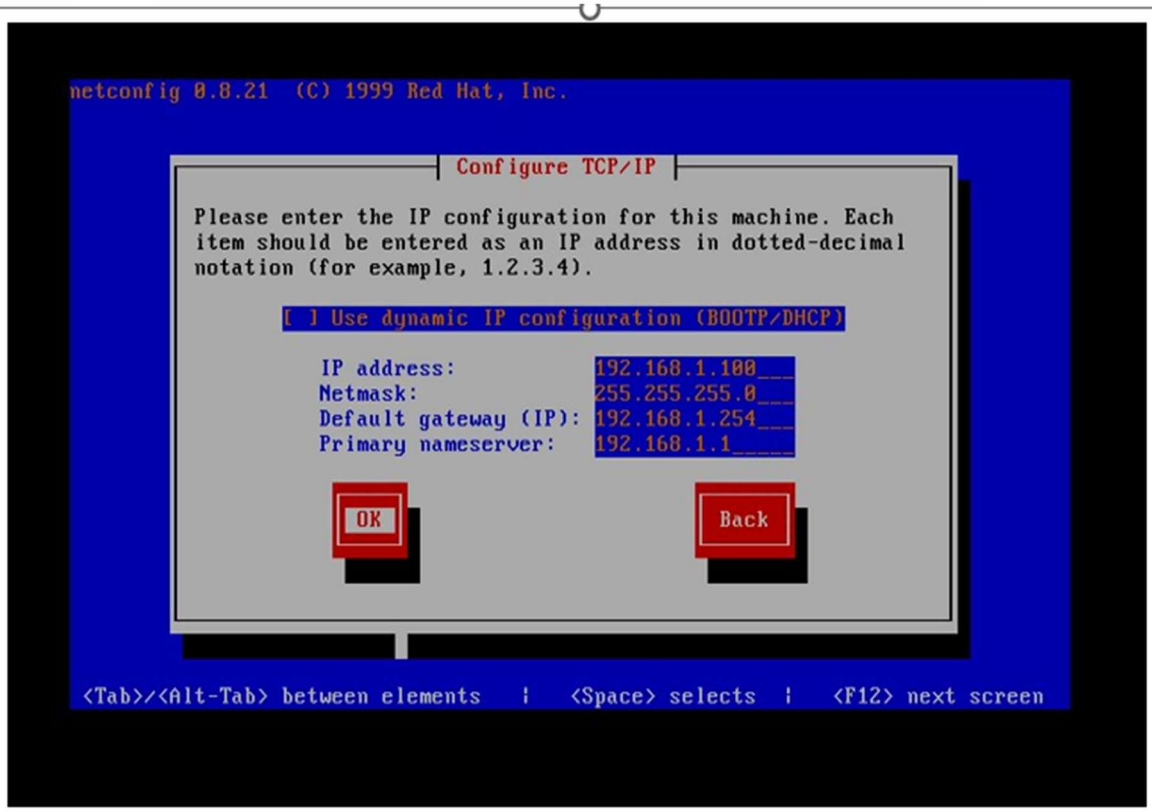


Fig.2. Snapshot

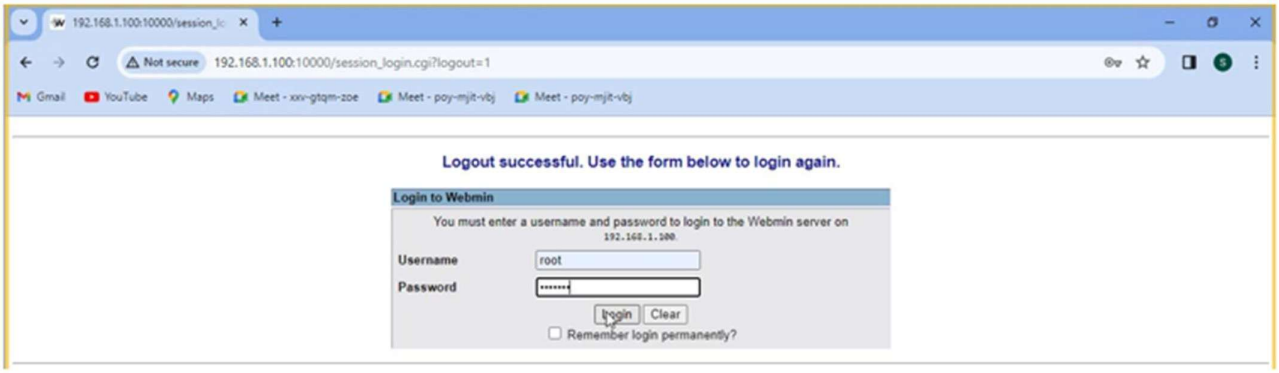


Fig.3. Insights

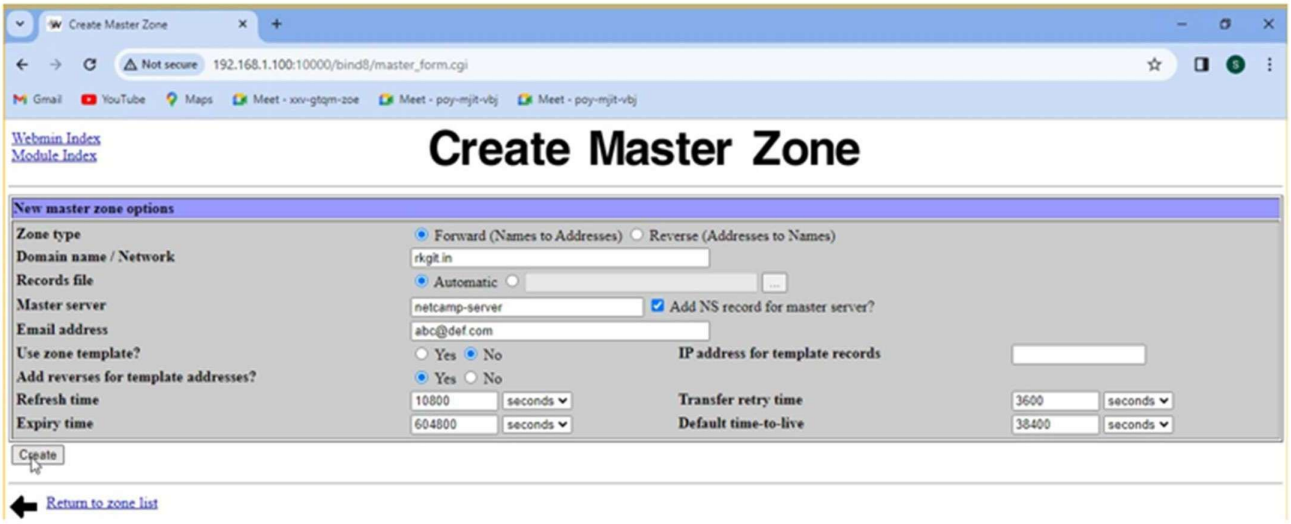


Fig.4. Dialog Box

Facilities required for proposed work

The successful development and deployment of the Linux-Based DNS Server and Domain Hosting System depend on appropriate hardware, software, and networking resources. These facilities ensure efficient configuration, testing, and hosting of multiple domains within a secure Linux environment.

Hardware Requirements

A workstation or personal computer with a multi-core processor (Intel i5/i7 or equivalent), 8 GB or higher RAM, and 500 GB SSD storage is essential to support server configurations and virtual environments. Stable internet connectivity and uninterrupted power supply are also required to maintain network communication during DNS and web server operations.

Software Requirements

The system uses Ubuntu Server or CentOS Linux as the primary operating system. The main software components include BIND9 for DNS setup, Apache2 for web hosting, and phpMyAdmin/MySQL for database management if needed. Supporting tools like VS Code, Nano, and Terminal utilities (ping, nslookup, dig) assist in configuration and testing.

Networking and Learning Resources

A reliable LAN or Wi-Fi network is necessary for DNS resolution and domain testing. Mentor guidance, online documentation, and Linux community resources enhance learning and troubleshooting efficiency.

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Category	Facilities
Hardware	Computer/Laptop (8–16 GB RAM, i5/i7 processor, 500 GB SSD), Internet connection
Software	Ubuntu/CentOS Linux OS, BIND9, Apache2, phpMyAdmin/MySQL, Visual Studio Code, Nano Editor
Programming Tools	Shell Command Utilities (ping, nslookup, dig), Configuration Files (named.conf, zone files)
Networking	Localhost Testing (127.0.0.1), LAN/Wi-Fi Connection for DNS Resolution
Human Resources	Mentor Guidance, Linux Tutorials, Official BIND & Apache Documentation, GitHub Repositories

Table 1: Facilities Required

Real world Application

The Linux-Based DNS Server and Domain Hosting System has several practical and industrial applications in today's technology-driven environment. It plays a crucial role in ensuring the smooth functioning of network communication, domain resolution, and website management across both private and public infrastructures.

In the corporate world, organizations utilize Linux-based DNS servers to manage internal and external domains, enabling efficient control over network resources and data traffic. This system ensures that users can access websites, intranet portals, and online services through easily memorable domain names instead of complex IP addresses. Educational institutions and research organizations also benefit by hosting their campus websites, student portals, and mail servers on self-managed DNS and hosting platforms, ensuring cost-effectiveness and customization.

Web hosting companies, Internet Service Providers (ISPs), and data centres use similar configurations to provide shared and dedicated hosting services to clients. The implementation of primary and secondary DNS servers ensures high availability and reliability, minimizing downtime and improving performance.

Moreover, the system can be integrated into cybersecurity training and ethical hacking environments, where students and professionals learn about network protocols, DNS spoofing prevention, and secure configuration practices. It also supports cloud-based and hybrid infrastructures, where scalable DNS services are vital for containerized and virtualized deployments.

Overall, the Linux-Based DNS Server and Domain Hosting System demonstrates how open-source technologies can be effectively used to build secure, reliable, and scalable network infrastructures, forming the backbone of modern digital communication and web hosting services.